AF-400 INVERTER DRIVES

INVERTER ASSEMBLY (400 KVA) - 331X401AAG01
(300 KVA) - G02

OPERATION - TROUBLE SHOOTING - REPAIR

GEK-62519

DIAGRAMS

INVERTER ASM.	ELEM	36C764880AA
	CONN	36D868885AA
PHASE MODULE	ELEM	36C764878AA
	CONN	36C764882AA
CONVERTER MODULE	ELEM	36C764879AA
	CONN	36C764887AA
COMM. POWER SUPPLY	ELEM	36C764881AA
	CONN	36D868889AA

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company

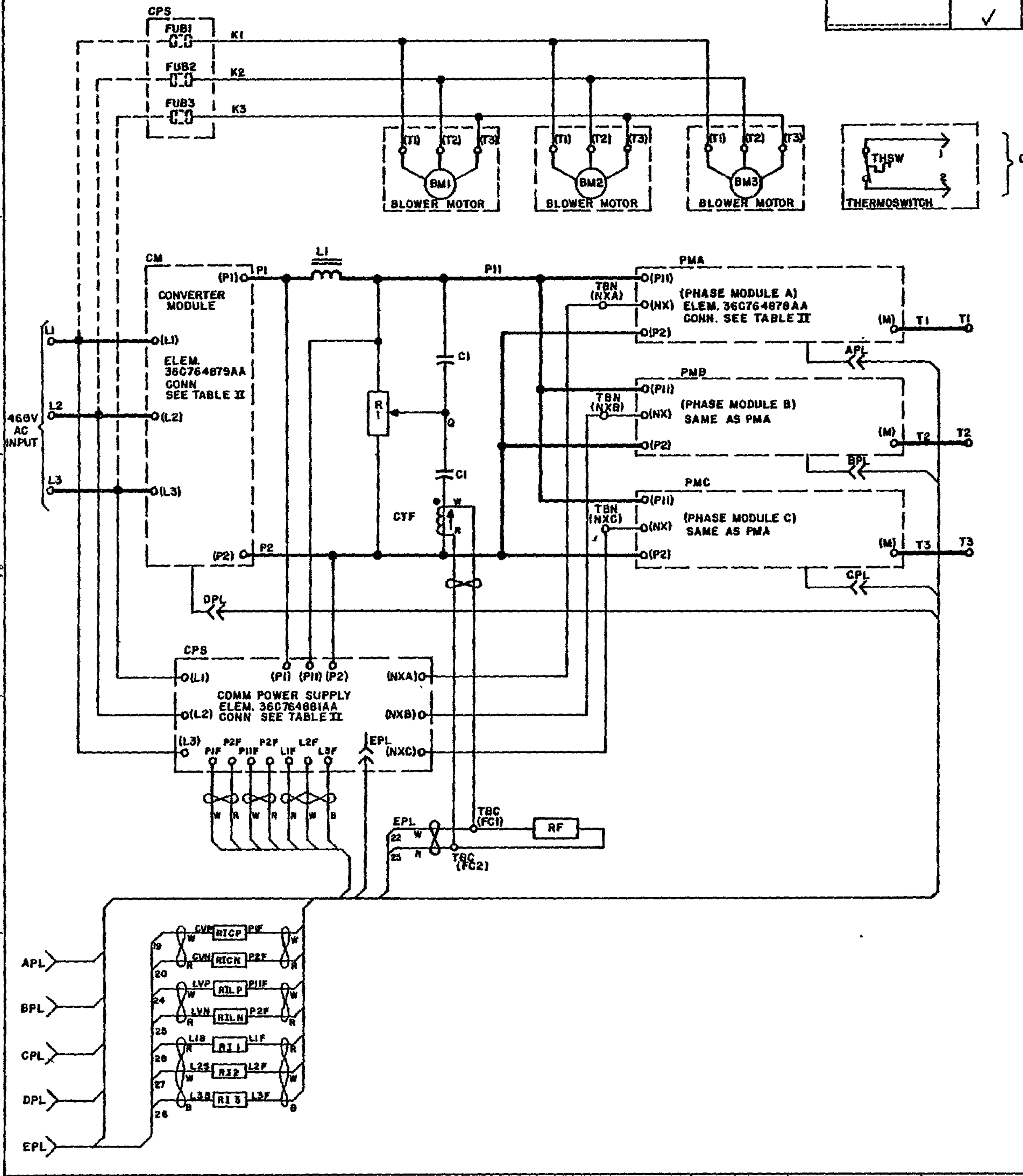
GENERAL ELECTRIC

demark of the General Electric Co.

GENERAL ELECTRIC 36 4880AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
AF-400 INVERTER ASMI
FIRST MADE FOR AF-400
331X401, 402, 403AA



NOTES:

1. THE ASSEMBLY CATALOG NUMBER GROUP DISTINCTION, MODULE CATALOG NUMBERS AND SOME COMPONENT VALUES ARE DEPENDENT UPON KVA RATING. TABLE I LISTS VARIABLE COMPONENT VALUES.
2. THE L1 CHOKE IS MOUNTED REMOTE TO THE INVERTER ASSEMBLY. ITS VALUE IS DEPENDENT UPON KVA RATING AND IS LISTED IN TABLE I.
3. ELEMENTARY WIRING SYMBOLS
   - W R —INDICATES TWISTED WIRES, COLOR INDICATED BY LETTER
   - 1 2 3 —INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG.
   - —QUICK DISCONNECT "STAB-ON" CONNECTION.
   - —DOT INDICATES POSITIVE TERMINAL WHEN CURRENT FLOWS IN THE DIRECTION OF THE ARROW.
4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
5. RESISTORS RIC, RIL, RI1, RI2, RI3 ARE RESISTANCE ISOLATOR RESISTORS, AND ARE MOUNTED AS AN INTEGRAL PART OF THE WIRE HARNESS
6. BM1, BM2, BM3 USED ON 300/400 KVA
   BM1, BM2 USED ON 200 KVA
   BM1 USED ON 100/50KVA
7. SEE TABLE II FOR CONNECTION DIAGRAMS.
8. TABLE I

KVA	331X	L1	C1//C1	C1	R1	RF
		μH	μF	μF	OHM/W	OHM
400	401AAG01	400	22400	44800	1000/800	8.3 4X33 2Ω IN PAR
300	401AAG02	530	16800	33600	1500/400	11.1 3X33.2Ω IN PAR.
200	402AAG01	800	11200	22400	1500/400	16.6 2X33 2Ω IN PAR
100	403AAG01	1600	5600	11200	2500/200	16.6 2X33 2Ω IN PAR
50	403AAG02	3200	2800	5600	2500/200	33.2Ω

TABLE II

KVA	PHASE MODULE	CONVERTER MODULE	INVERTER ASSEMBLY	COMM. P/S
400/300	36C764882AA	36C764887AA	36D868885AA	36D868889AA
200/	36C764883AA	36C764888AA	36D868886AA	36D868890AA
100/50	36C764884AA	36C764889AA	36D868887AA	36D868891AA

REVISIONS				PRINTS TO	
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department	1.	May 20, 1977		5B(2)22	5(S)
				5QC(BK)5	5D(BK)
				5(TS)50	5L(BW)5
				AW(BW)	5AE(BW)
				DS(REP)	5QC(B)

MADE BY M.S. Pugh AUG 17, 1976
CHECKED Jan 31, 1977
APPROVALS
SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION
36C764880AA
CONT ON SHEET SH NO.

REV NO. 1

FF-509 F (9 75)
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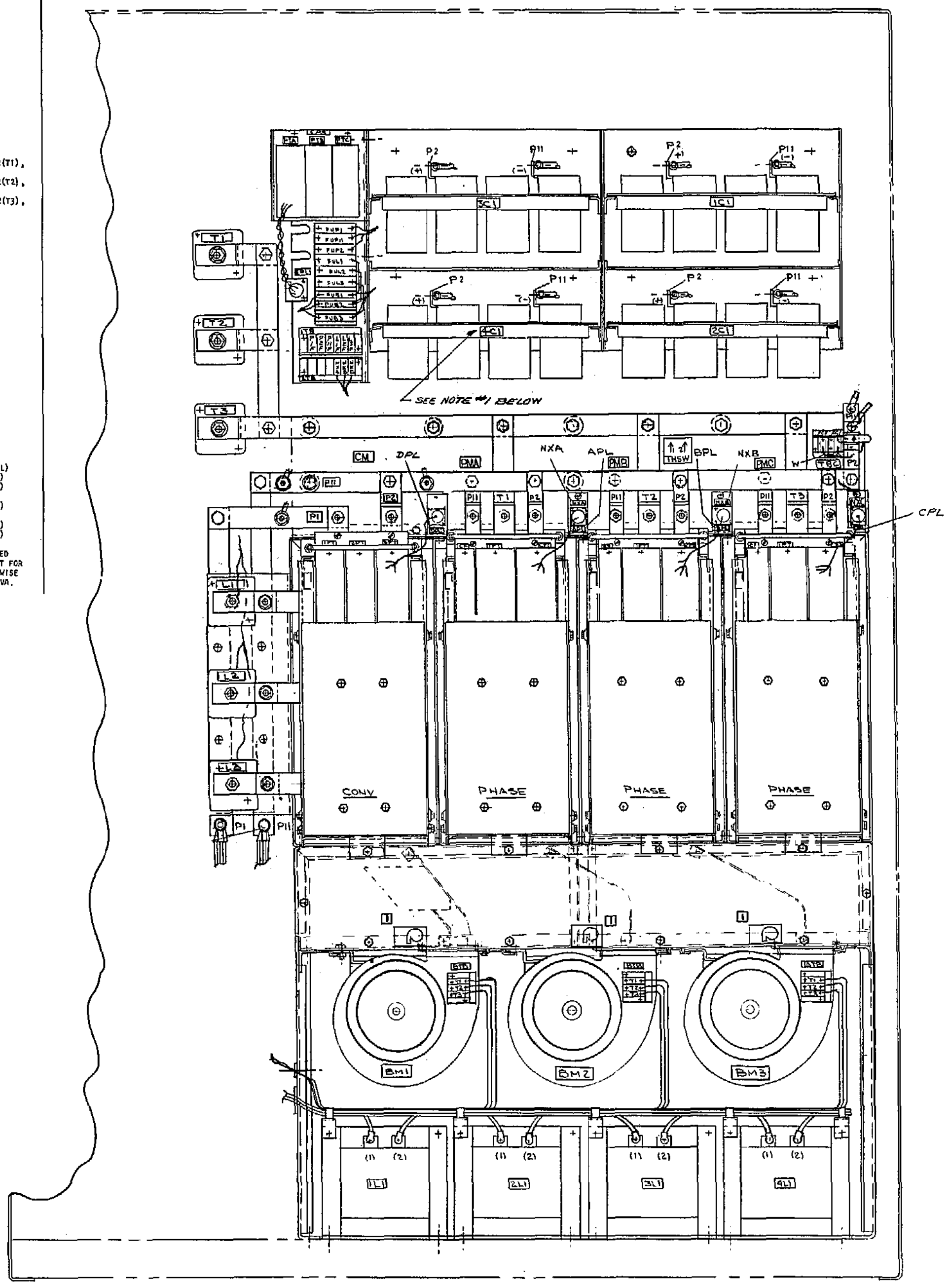
INVERTER ASM WIRING
FOR 400 KVA AND FOR
300 KVA SEE NOTE BELOW(✱)

WIRE NO.	CONNECTION
	BUS BAR - 1/4" X 1-1/2"
P1	BUS(P1),CM(P1)
P2	CM(P2),PMA(P2),PMB(P2), PMC(P2)
P11.	BUS(P11),PMA(P11),PMB(P11), PMC(P11)
L1	CM(L1),PTB(L1)
L2	CM(L2),PTB(L2)
L3	CM(L3),PTB(L3)
T1	PMA(M),PTB(T1)
T2	PMB(M),PTB(T2)
T3	PMC(M),PTB(T3)
	#4 BLK-SILICONE
P1	1L1(1),BUS(P1)
P1	2L1(1),BUS(P1)
P1	3L1(1),BUS(P1)
✱P1	4L1(1),BUS(P1)
P11	1L1(2),BUS(P11)
P11	2L1(2),BUS(P11)
P11	3L1(2),BUS(P11)
✱P11	4L1(2),BUS(P11)
	#6 BLK-SILICONE
P2	1C1(P2),BUS(P2)
P2	2C1(P2),BUS(P2)
P2	3C1(P2),BUS(P2)
✱P2	4C1(P2),BUS(P2)
P11	1C1(P11),BUS(P11)
P11	2C1(P11),BUS(P11)
P11	3C1(P11),BUS(P11)
✱P11	4C1(P11),BUS(P11)
	#14 BLK
L1	BUS(L1), FUB1(2)
L2	BUS(L2), FUB2(2)
L3	BUS(L3), FUB3(2)
K1	FUB1(1), BM1(T1), BM2(T1), BM3(T1)
K2	FUB2(1), BM1(T2), BM2(T2), BM3(T2)
K3	FUB3(1), BM1(T3), BM2(T3), BM3(T3)
NXA	2TB(NXA), TBN(NXA)
NXB	2TB(NXB), TBN(NXB)
NXC	2TB(NXC), TBN(NXC)
P1	BUS(P1), FUP1(2)
P2	BUS(P2), FUP2(2)
P11	BUS(P3), FUP11(2)
	RESISTOR WIRES
P1F0W	1TB(P1F), EPL(19)
P2F0R	1TB(P2F), EPL(20)
P11F0W	1TB(P11F), EPL(24)
P2F0R	1TB(P2F), EPL(25)
L1F0R	1TB(L1F), EPL(28)
L2F0W	1TB(L2F), EPL(27)
L3F0B	1TB(L3F), EPL(26)
	PLUG WIRING
APL	DRIVER (APL), PMA(APL)
BPL	DRIVER(BPL), PMB(BPL)
CPL	DRIVER(CPL), PMC(CPL)
DPL	DRIVER(DPL), CM(DPL)
EPL	DRIVER(EPL), CPS(EPL)
EPL22 0W	DRIVER(EPL), TBC(FC1)
EPL23 0R	DRIVER(EPL), TBC(FC2)
✱	NOTE: WIRE NO. MARKED WITH (✱) OMIT FOR 300 KVA OTHERWISE SAME AS 400 KVA.



NOTE #1: 4L1 AND 4C1 NOT USED
ON 300 KVA INVERTER.

GENERAL ELECTRIC

36D868885AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-				
APPLIED PRACTICES.	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	+ −	+ −	+ −

KEY NO 0/2

CONT ON SHEET SH NO.

TITLE INVERTER
CONNECTION DIAG

FIRST MADE FOR AF400

FOR 400 AND 300 KVA

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS

1 Aug 23 '77 SP867
CHGS LABEL LOCATIONS
CHGS BLOWER MOUNTING LOCATIONS
CHGS WIRING HARNESS

2 JAN 30, 1978 SV-872 #21
CHG RESISTOR WIRE
IN WIRE TABLE

PRINTS TO	
5B(B)M	5J(BW)
5R(2)BW	
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

Dec 29, 1976
Mar 31, 1977

APPROVALS

SP VARIATOR DIV OR DEPT
ERIE LOCATION

36D868885AA
CONT ON SHEET SH NO.

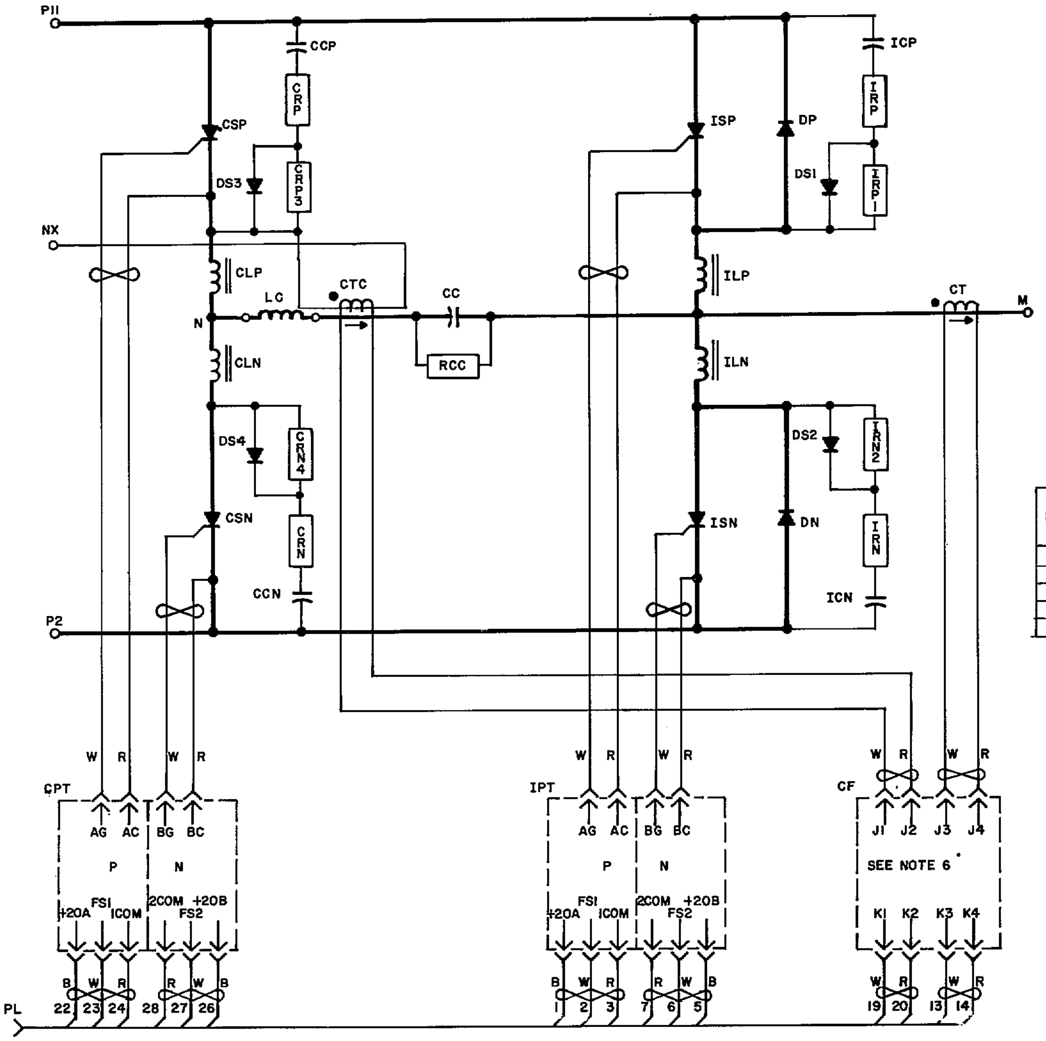
GENERAL ELECTRIC 36C764878AA

CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:–				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
AF-400 PHASE MODULE

FIRST MADE FOR AF-400
331X411, 412, 413AA



NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES OF I OR C TO INDICATE INVERTER OR COMMUTATION SECTION. SUFFIXES OF P OR N INDICATE POSITIVE OR NEGATIVE. THE BASIC NOMENCLATURE IS:
   - C — CAPACITOR
   - D — DIODE
   - L — CHOKE
   - PT — PULSE TRANSFORMER PRINTED CIRCUIT CARD
   - PL — CONNECTOR PLUG
   - R — RESISTOR
   - S — SCR
   - DS — SNUBBER DIODE
2. PULSE TRANSFORMER CARD CIRCUITRY IS NOT SHOWN FOR SIMPLICATION. FOR DETAILS REFER TO THE INSTRUCTION BOOK.
3. ELEMENTARY WIRING SYMBOLS
   - R W B —INDICATES TWISTED WIRES, COLOR INDICATED BY LETTER.
   - 2 3 —INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG.
   - —QUICK DISCONNECT "STAB-ON" CONNECTION.
   - —DOT INDICATES POSITIVE TERMINAL WHEN CURRENT FLOWS IN THE DIRECTION OF THE ARROW.
4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
5.

KVA	331X	CC	LC	IRP, CRP IRN, CRN	IRP1, CRP3 IRN2, CRN4	ICP, CCP ICN, CCN	ILP ILN	CLP CLN	RCC
		μF	μH	OHM/W	OHM/W	μF	SAT μH	SAT μH	K OHM /W
400	411AAG01	80	5	2.5/40	20/40	1	1.5	1.5	280/4
300	411AAG02	60	7.5	2.5/40	20/40	1	2	2	280/4
200	412AAG01	40	10	2.5/40	20/40	1	2.6	2.6	560/2
100	413AAG01	20	25	10/40	20/40	.25	4	4	560/2
50	413AAG02	10	53	10/40	20/40	.25	8	8	1000/2

6. THE CURRENT FEEDBACK CARD WILL CHANGE DEPENDING ON KVA RATING. GROUP OF CARD AND JUMPER SELECTION FOR NOMINAL CURRENT ARE–

KVA	CF 193X382AA	NOMINAL INV. CURRENT A RMS	JUMPER SELECTION	NOMINAL COMM. CURRENT A PEAK
400	G05	580 500 420	XC XB XA	2100
300	G04	420 350 280	XC XB XA	1500
200	G03	280 210 175	XC XB XA	1000
100	G02	140 105 84	XC XB XA	500
50	G01	70 56 42	XC XB XA	250

DRAWING NO. CONT ON SHEET SH NO.

REV NO. 1

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS			
1	May 20, 19[illegible] J. Binder	[illegible]	[illegible]

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

MADE BY M D Porter AUG. 16, 1976
ISSUED [illegible] Jan. 31 1977
APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764878AA
CONT ON SHEET SH NO.

FF-809-F (9-75)
PRINTED IN U.S.A.

GENERAL ELECTRIC 36C764882AA

CONT ON SHEET SH NO.

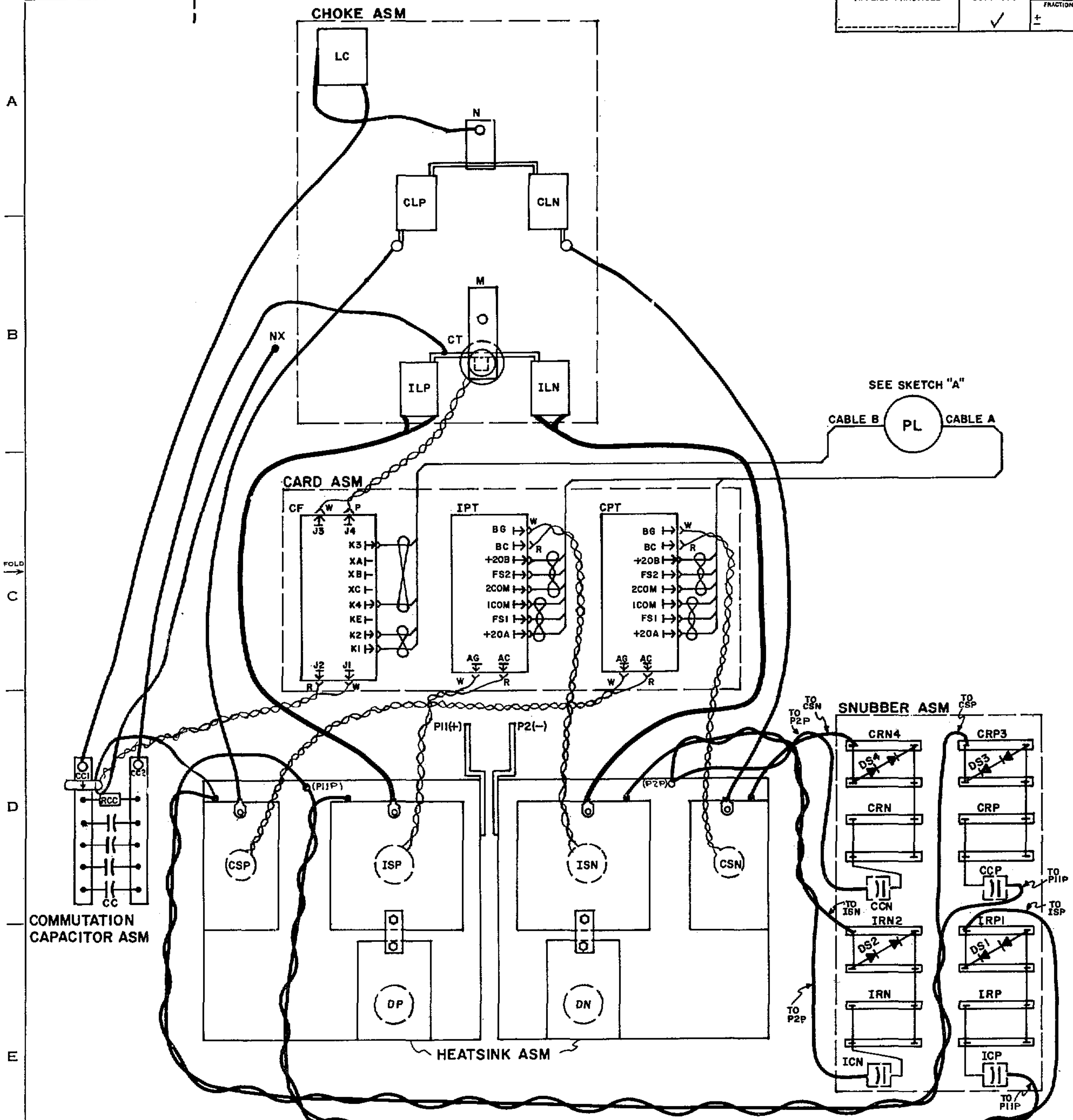
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE **CONNECTION DIAGRAM**
400/300 KVA PHASE MODULE

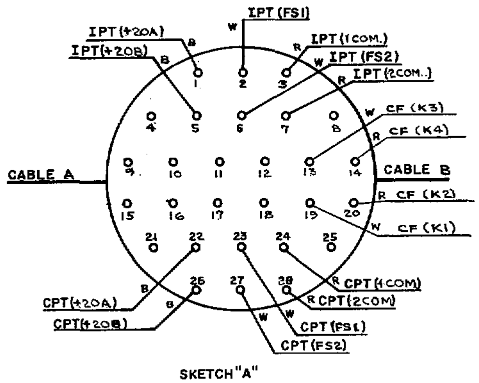
FIRST MADE FOR AF-400

331X411AAG01

REF TO WIRE CHART
36A338012XBG01



VIEW FROM WIRE SIDE OF PLUG SOCKET



SKETCH "A"

NOTES.

1. ALL POWER WIRING IS EITHER BUS OR CONNECTORS PER MECHANICAL ASSEMBLY DRAWING.
2. WIRES ARE MARKED AS INDICATED. AC, AG ; BC & BG ARE TWISTED PAIRS OF #20 AWG RED & WHITE. GATES (G) ARE WHITES.
   TWISTED RED, WHITE AND BLUE WIRES, TO PULSE TRANSFORMER CARDS, ARE #20 WIRES
   TWISTED RED & WHITE WIRES TO CURRENT FEEDBACK CARD ARE #20 AWG.
3. ALL OTHER CONTROL WIRES ARE #16 AWG RED, EXCEPT AS OTHERWISE INDICATED.

REV NO.
0
1

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

W.G. Bender	FEB 25, 1977	RECORD CHG

P2P WAS P2, P1IP WAS P1I
CHGD CONN. FROM LC — M TO LC — N, CHGD CONN FROM CC2 — N TO CC2 — M
CHGD SK. "A"

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

MADE BY M B Pozla MAY 20, 1976
ISSUED MAY 26, 1976

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764882AA
CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

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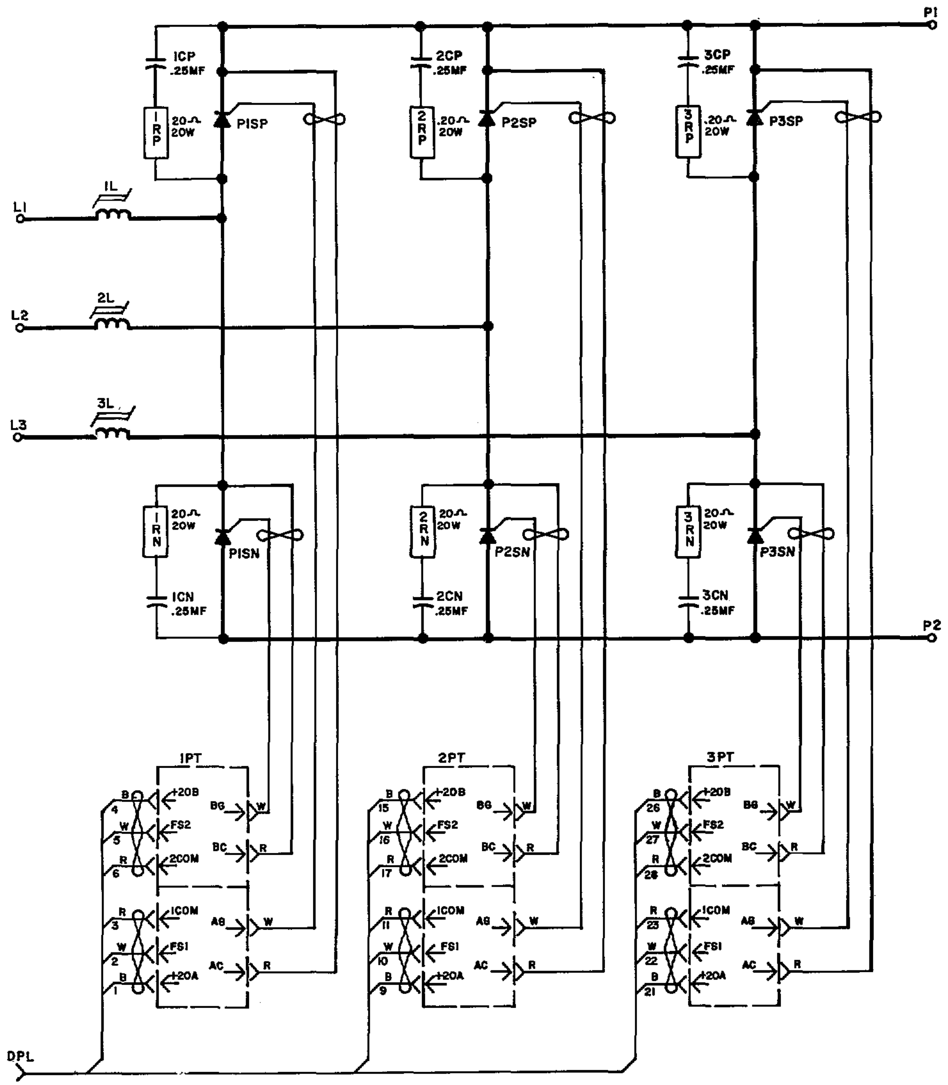
GENERAL ELECTRIC

36C764879AA

CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	√	±	±	±

TITLE ELEMENTARY DIAGRAM
AF-400 CONVERTER MODULE
FIRST MADE FOR STANDARD LINE
331X421,422,423AA



NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES P1, P2 ETC., TO INDICATE *PHASE CONTROL AND PHASE.* SUFFIXES OF P OR N INDICATE POSITIVE OR NEGATIVE. BASIC NOMENCLATURE IS:
   - C — CAPACITOR
   - D — DIODE
   - L — CHOKE
   - PT — PULSE TRANSFORMER PRINTED CIRCUIT CARD
   - PL — CONNECTOR PLUG
   - R — RESISTOR
   - S — SCR
   - X — AUXILIARY SCR
2. PULSE TRANSFORMER CARD CIRCUITRY IS NOT SHOWN FOR SIMPLIFICATION. FOR DETAILS REFER TO THE INSTRUCTION BOOK.
3. ELEMENTARY WIRING SYMBOLS
   - R W B — INDICATES TWISTED WIRES, COLORS INDICATED BY LETTER
   - 1 2 3 — INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG PL.
   - →≫— — QUICK DISCONNECT "STAB ON" CONNECTION
4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
5. 1L, 2L, 3L; 3 FERRITE CORES 1 TURN

DRAWING NO.
CONT ON SHEET SH NO.

REV NO.
0

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

MADE BY M D Pozyk AUG. 24, 1976
ISSUED D. E. Williams Jan. 31 1977
APPROVALS EAS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764879AA
CONT ON SHEET SH NO.

GENERAL ELECTRIC | 36C764887AA | CONT ON SHEET | SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

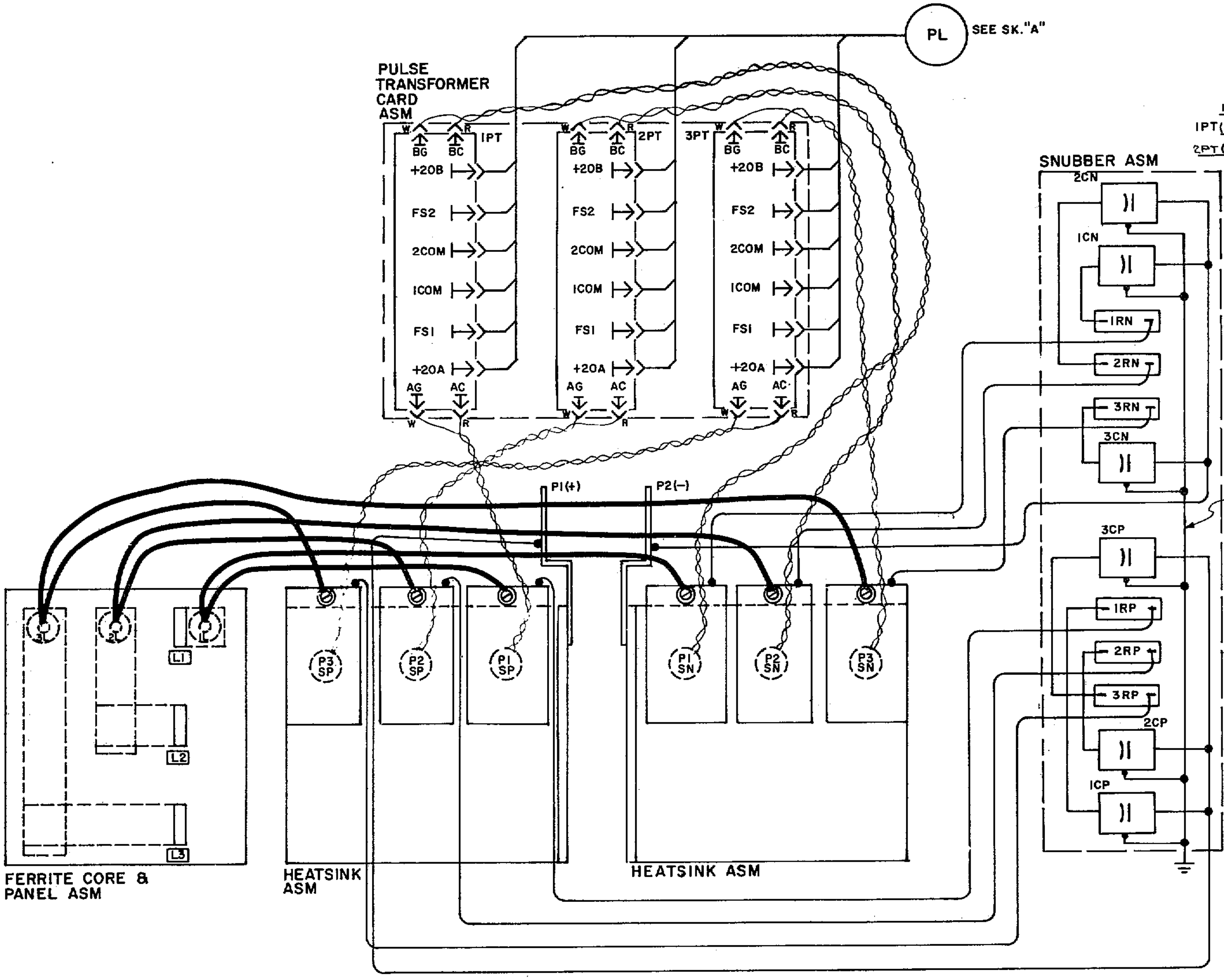
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	√	±	±	±

TITLE CONNECTION DIAGRAM
400 / 300 KVA ~~CONVERTER~~ MODULE

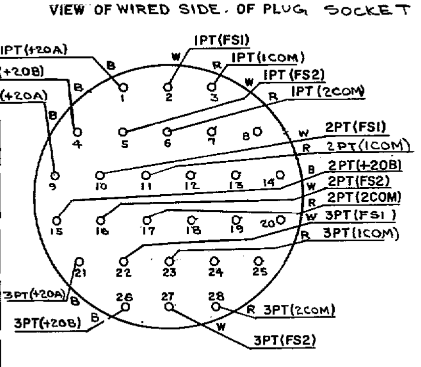
FIRST MADE FOR AF-400

331X421AAG01

REF TO WIRE CHART 36A358012XC G01



VIEW OF WIRED SIDE. OF PLUG SOCKET



SKETCH "A"

NOTES:

1. ALL POWER WIRING IS EITHER BUS OR CONNECTORS PER MECHANICAL ASSEMBLY DRAWINGS.
2. WIRES ARE MARKED AS INDICATED.
   AC, AG, BC, BG ARE TWISTED PAIRS OF # 20AWG RED AND WHITE.
   GATES (G) ARE WHITE.

   TWISTED RED, WHITE AND BLUE WIRES TO PULSE TRANSFORMER CARDS ARE # 20 WIRE.
3. ALL OTHER CONTROL WIRES ARE # 16 AWG RED EXCEPT AS OTHERWISE INDICATED.

DRAWING NO. | CONT ON SHEET | SH NO.

REV NO. ~~0~~ ~~1~~ 2

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS

1	M. B. Payha SEPT.22 1976	SV-867
2	FEB 25 1977 J. Bender	RECORD CHG
	CHG'D SK A TO WIRE SIDE VIEW	

PRINTS TO

5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW
DS(REP)	5QC(B)

MADE BY M B Payha MAY 12, 1976
ISSUED MAY 26 1976

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764887AA
CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

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GENERAL ELECTRIC

36C764881AA

CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
COMMUTATION POWER SUPPLY

FIRST MADE FOR AF-400

331X431AAG01

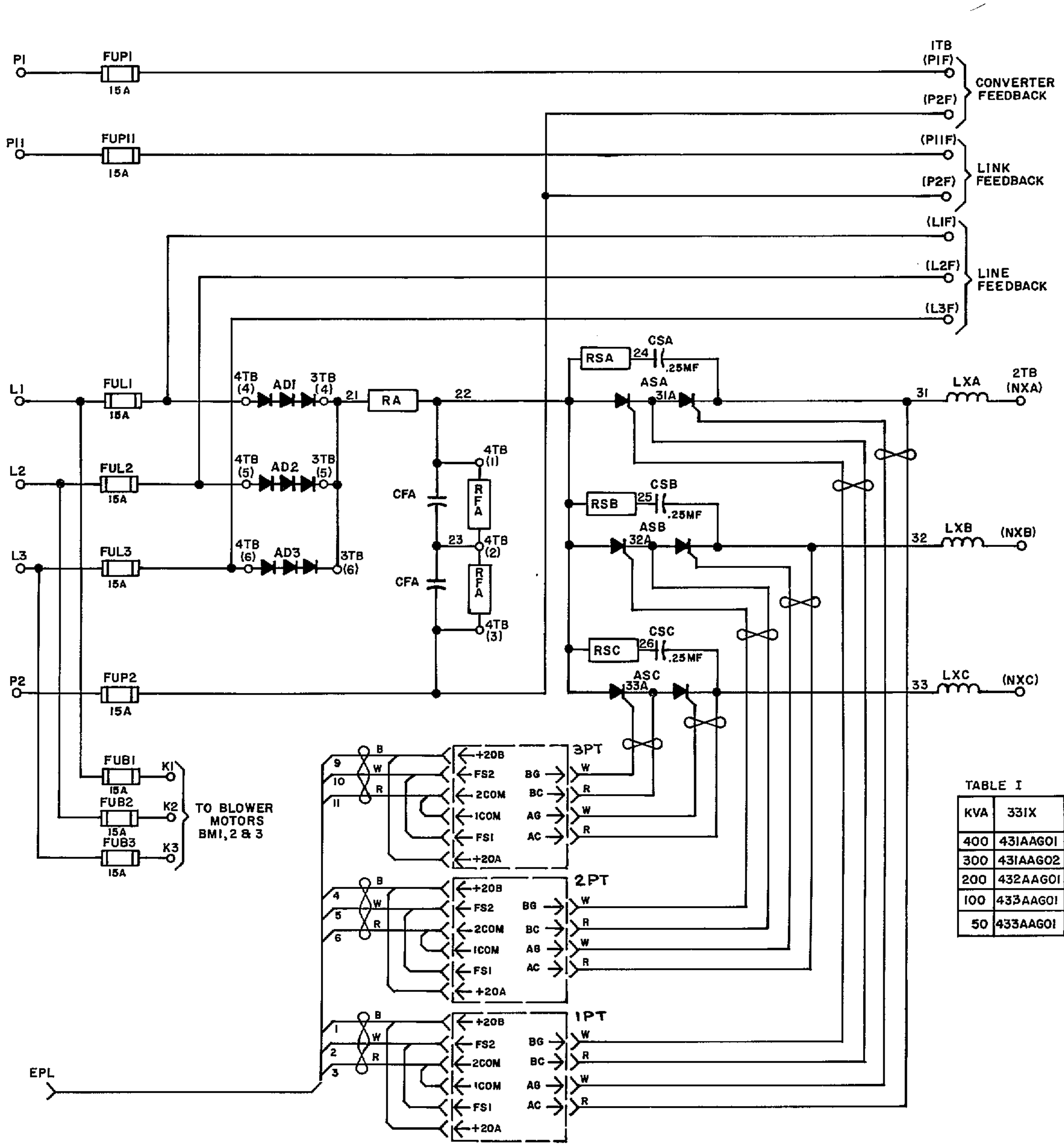


TABLE I

KVA	331X	LX µH	CFA µF	RA OHMS/W	RFA OHM/CONN/W	RSA OHM/CONN/W
400	431AAG01	54	1400	5/100	10K/2X5K SER/20	100/2 X 200 PAR/40
300	431AAG02	54	1400	5/100	"	"
200	432AAG01	110	850	10/50	"	200/2X100 SER/40
100	433AAG01	220	850	10/50	"	400/2X200 SER/40
50	433AAG01	"	850	10/50	"	"

REV NO.

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1	Feb. 1, 1977 E Gualtieri DW	SV872 #12A
2	NOV. 7, 1977 O.R. Baker	PER #13 SV-872 #13A
3	JAN 31, 1978	SV872 #20
4	M. Crotty June 21, '78	PER SV-876 #8

CHG MARKING
1PT WAS PTA
2PT WAS PTB
3PT WAS PTC

PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

MADE BY M J Pozzka AUG 25, 1976
M. Kazanauskas OCT 18, 1976

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764881AA
CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FF-809-F (9-73)
PRINTED IN U.S.A.

GENERAL ELECTRIC 36D868889AA

TITLE CONNECTION DIAGRAM

COMMUTATION POWER SUPPLY

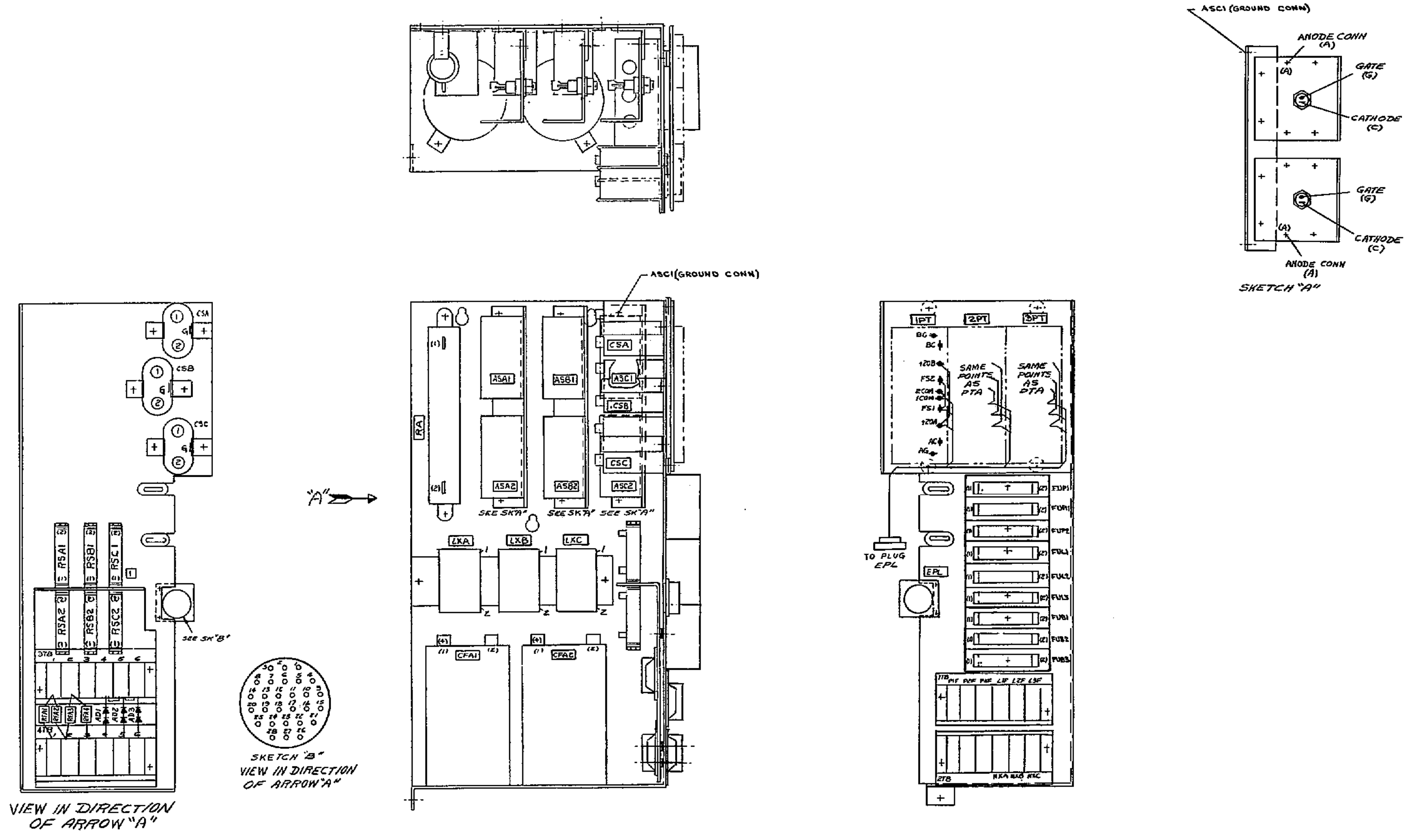
FIRST MADE FOR AF400

300/400 KVA 331X431AAG1

REV NO. 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:–

APPLIED PRACTICES	SURFACES	FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±



SKETCH "A"

SKETCH "B"
VIEW IN DIRECTION OF ARROW "A"

VIEW IN DIRECTION OF ARROW "A"

WIRE TABLE

REV	WIRE NO.	CONNECTION
		#16 BLACK POWER WIRING
	P1F	FUP1(1),1TB(P1F)
	P2F	1TB(P2F),FUP2(1),CFA2(2),4TB(3)
	P11F	1TB(P11F),FUP11(1)
	L1F	1TB(L1F),FUL1(1),4TB(4)
	L2F	1TB(L2F),FUL2(1),4TB(5)
	L3F	1TB(L3F),FUL3(1),4TB(6)
	21	3TB(4)*,3TB(5)*,3TB(6)*,RA(2)
	22	4TB(1),CFA1(1),ASC1(A),ASB1(A),ASA1(A),RA(1)
1	22	ASA1(A),RSC2(1),RSC1(2),RSB1(2),RSB2(1)
1	22	RSB1(2),RSA1(2),RSA2(1)
	23	4TB(2),CFA2(1),CFA1(2)
	24	RSA2(2),RSA1(1),CSA(1)
1	25	RSB2(2),RSB1(1),CSB(1)
	26	RSC2(2),RSC1(1),CSC(1)
1	31	CSA(2),LXA(1),ASA2(C)▲
	32	CSB(2),LXB(1),ASB2(C)▲
	33	CSC(2),LXC(1),ASC2(C)▲
	NXA	2TB(NXA),LXA(2)
	NXB	2TB(NXB),LXB(2)
	NXC	2TB(NXC),LXC(2)
	31A	ASA1(C)*ASA2(A)
1	32A	ASB1(C)*ASB2(A)
	33A	ASC1(C)*ASC2(A)
1	G	CSC(G),CSB(G),CSA(G),ASC1(G)
		* METAL JUMPER CONNECTION
	L1	FUL1(2),FUB1(2)
	L2	FUL2(2),FUB2(2)
	L3	FUL3(2),FUB3(2)
		#20 RED & WHITE TWISTED SCR GATE LEAD WIRING
	W	ASA1(G),1PT(BG)
	R	ASA1(C),1PT(BC)
	W	ASA2(G),1PT(AG)
	R	ASA2(C),1PT(AC)
	W	ASB1(G),2PT(BG)
1	R	ASB1(C),2PT(BC)
	W	ASB2(G),2PT(AG)
	R	ASB2(C),2PT(AC)
	W	ASC1(G),3PT(BG)
	R	ASC1(C),2PT(BC)
	W	ASC2(G),3PT(AG)
	R	ASC2(C),3PT(AC)
		#20 BLUE, WHITE & RED TWISTED PLUG WIRING (EPL)
	B	EPL(1),1PT(+20B),1PT(+20A)**
	W	EPL(2),1PT(FS2),1PT(FS1)**
	R	EPL(3),1PT(2COM),1PT(1COM)**
	B	EPL(4),2PT(+20B),2PT(+20A)**
	W	EPL(5),2PT(FS2),2PT(FS1)**
	R	EPL(6),2PT(2COM),2PT(1COM)**
	B	EPL(9),3PT(+20B),3PT(+20A)**
	W	EPL(10),3PT(FS2),3PT(FS1)**
	R	EPL(11),3PT(2COM),3PT(1COM)**
		** LAST CONNECTION POINT NOT TWISTED
1		▲ SOLDER BARE END OF THE #16 BLACK WIRE WITH THE BARE END OF THE #20 RED WIRE, TO THE CATHODE (C) OF THE SCR TOGETHER AS ONE OPERATION.

REVISIONS

2	Jan 31, 1978	SVB72 #20	CHG LABEL MARKING ON ASM. 1PT WAS PTA 2PT WAS PTB 3PT WAS PTC. GATE LEAD WIRE WERE PTA, PTB, PTC. TWISTED PLUG WIRING WAS PTA, PTB, PTC
1	June 13, 1977	PCR AN 55-7-1428	CHGD WIRE NO 22 WIRING TO AGREE WITH CHGD RES MTG ADDED "G" WIRE CONN. ADDED ▲ NOTE FOR WIRE TABLE CONN.

PRINTS TO

5B(8)M	SJ(BW)
5R(2)BW	
5L(2)BW	
AW(BW)	
5D(CO)	
JA(CO)	

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

Sept 23, 1976
Oct 18, 1976

APPROVALS

SPEED VARIATOR DIV OR DEPT

ERIE, PA. LOCATION

36D868889AA CONT ON SHEET SH NO